Shiwen LIU

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Education

University of Bristol (UK), Bristol, UK **Sep 2024 – Dec 2025**

MSc in Cyber Security (Infrastructures Security)

- Expected to graduate with Merit
- *Part of 3+1 AMD Cooperative Program with NJUST*

Nanjing University of Science and Technology, Nanjing, China **Sep 2021 – Oct 2025**

BEng in Cyberspace Security

- GPA: 3.46/4.0
- Core Courses: AI with Information Society *(99), Cybersecurity Course Design(Ⅲ) – Deep Learning (90), Cyberspace Technology(Ⅲ) - Artificial Intelligence (90),Python Program (90)*

Publication

Large Language Model --- MirrorMind

"MirrorMind: A Dual-Agent AI Interview Simulator Powered by LLMs." Accepted *at **ADMA 2025.***

Intrusion Detection System --- BERTector

"BERTector: An Intrusion Detection Framework Constructed via Joint-dataset Learning Based on Language Model." arXiv preprint arXiv:2508.10327 (2025).

Intrusion Detection System --- ECIDS

"Classifier Follows Embedder: A Lightweight Semantic Packet-level Intrusion Detection System for Real-time Devices."

Research Experience

Protocol-Specific Packet-Based Intelligent Intrusion Detection ***Feb 2025 – Aug 2025***

- Designed and implemented a packet-level intrusion detection system for wireless communication.
- Proposed a two-stage architecture combining a Word2Vec-based **embedder** and a lightweight **classifier**.
- Evaluated on multiple datasets including collected simulated datasets of mavlink protocol and public datasets such as CICIDS2017 and UNSW-NB15.

A Dual-Agent AI Interview Simulator Powered by LLMs ***Dec 2024 – Mar 2025***

- Developed a comprehensive interview assistance system that leverages LLMs to generate customized interview questions and simulate realistic candidate responses for effective interview preparation.
- Integrated multimodal technology, including OCR for resume text extraction and advanced prompt engineering, can dynamically adjust interview conversations based on user profiles.
- Anonymous code repository: *MirrorMind-CE8D*

LLM-Based Feature Selection for Drug Registration ***Nov 2024 – Feb 2025***

- Leveraged large language models to select latent features from regulatory texts with minimal labeled data
- Reduced reliance on large datasets through non-data-driven semantic analysis
- Resulted in a **patent** for the novel feature extraction approach

Prompt-Based Event Extraction Platform ***Undergraduate Research, 2024***

- Developed platform using PAIE (Prompting Argument Interaction for Event Argument Extraction) method to identify multi-trigger, multi-role events with higher accuracy
- Employed In-Context Learning (ICL) to reduce false positives and improve role classification.
- Awarded a **software copyrigh**t for the final system

Additional Experience

- Conducted multiple ICS & IoT evaluation assessments, including traffic-inspection deployment, controlled system access testing, and network compliance reviews.
- Engaged in real-world incident modeling and mitigating, examining how adversarial methods could influence infrastructures under simulated conditions.
- Adversarial Robustness & Gradient-Based Image Recovery (Lab): Investigated robustness testing for image classifiers and analyzed gradient-based partial reconstruction as a research probe into model behavior and privacy characteristics.

Skills

Programming: Python (PyTorch, NumPy, Transformers), C/C++ (basic), PyQt

Security Tools: Snort, Wireshark, Metasploit, Burp Suite, Nmap

Machine Learning: PyTorch, Hugging Face Transformers, adversarial attack frameworks

Operating Systems: Linux (Ubuntu, Kali), Windows